Example 8.25 If $P(\mathrm{A}) = 0.37$, $P(\mathrm{B}) = 0.42$, $P(A \cap B) = 0.09$ then find $P(A \cup B)$.

Solution $P(\mathrm{A}) = 0.37$, $P(\mathrm{B}) = 0.42$, $P(A \cap B) = 0.09$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = 0.37 + 0.42 - 0.09 = 0.7$$

Example 8.26 A flower is selected at random from a basket containing 80 yellow, 70 red and 50 white flowers. Find the probability of selecting a yellow or red flower?

Solution:

Total number of flowers $n(S) = 80 + 70 + 50 = 200$

No. of yellow flowers $n(Y) = 80 \therefore P(Y) = \dfrac{n(Y)}{n(S)} = \dfrac{80}{200}$

No. of red flowers $n(R) = 70 \therefore P(R) = \dfrac{n(R)}{n(S)} = \dfrac{70}{200}$

Thinking Corner

$P(A \cup B) + P(A \cap B)$ is _____.

Y and R are mutually exclusive $P(Y \cup R) = P(Y) + P(R)$

Probability of drawing either a yellow or red flower

$$P(Y \cup R) = \frac{80}{200} + \frac{70}{200} = \frac{150}{200} = \frac{3}{4}$$

Example 8.27 Two dice are rolled together. Find the probability of getting a doublet or sum of faces as 4.

Solution When two dice are rolled together, there will be $6 \times 6 = 36$ outcomes. Let S be the sample space. Then $n(S) = 36$

Let A be the event of getting a doublet and B be the event of getting face sum 4.

Then $A = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$

$B = \{(1,3),(2,2),(3,1)\}$

$\therefore \quad A \cap B = \{(2,2)\}$

Then, $n(A) = 6$, $n(B) = 3$, $n(A \cap B) = 1$.

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{3}{36}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{36}$$

$\therefore P$ (getting a doublet or a total of 4) = $P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{6}{36} + \frac{3}{36} - \frac{1}{36} = \frac{8}{36} = \frac{2}{9}$$

Hence, the required probability is $\dfrac{2}{9}$.

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